## POWER ENTRY

J1
VBUS
USB_CC1
USB_CC2
USB_DM_C
USB_DM_P
USB_DP_C
USB_DP_P
GCT_USB4105-GF-A
R_CC1 5.1k
R_CC2 5.1k
F1 VBUS
MF-NSMF050-2
VBUS_F
D1 PTVS6V0S1UR
Q1 Q1_GATE SI2301CDS-T1-GE3
R3 100k
C1 10uF
+5V
GND
SHIELD

USB-C VBUS 5 V max 5.25 V. Order: fuse (2.5 A hold) → TVS (5.0 V standoff) → P-FET reverse stage.
CC1 and CC2 each 5.1k to ground: sink, 5 V only.

## USB INPUT

R25 22
R26 22
USB_DP_C
USB_DP_F
USB_DM_C
USB_DM_F

D+ and D- from the receptacle through 22 ohm series resistors to PA12 and PA11.

## MCU

+3V3_MCU +3V3_MCU
U5
STM32F103C8T6
+3V3_MCU +3V3_MCU +3V3_MCU +3V3_MCU
NRST
BOOT0
PC13
PC14
PC15
PD0
PD1
CFG_SCK
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PA0
PA1
PA2
PA3
PA4
PA5
PA8
PA9
PA10
USB_DM
USB_DP
PA13
PA14
PA15
CFG_SS_B
CFG_SCK
CFG_SDO
CFG_SDI
USB_DM
USB_DP
SWDIO
SWCLK
VSS
VSSA
GND
J4
SWDIO
SWCLK
NRST
GND
(no value)
+3V3
R21 10k
R22 10k
SW1 PTS645SM43SMTR92 LFS
3T/4B0000FYLBC
Y1
PC14
PC15
+3V3_MCU
C27 100nF
C28 100nF
C29 100nF
C30 100nF
C31 100nF
C32 4.7uF
C33 100nF
C34 10pF
C35 10pF
FB2
(no value)
+3V3_MCU

STM32F103C8T6 on 3V3. Streams the bitstream to flash U2 over SPI (138,183 byte bitstream > 64 KB MCU flash).
8 MHz HC49 3T/4B0000FYLBC, CL 8 pF, ESR 150 ohm max, C0 5 pF max; assumed stray 3 pF: C = 2 x (8 − 3) = 10 pF each.
SWD header: 3V3, SWDIO, SWCLK, NRST, GND.

## 3V3 LDO

U3
AMS1117-3.3
+5V
+3V3
C2 10uF
C3 10uF
C4 10uF
C5 10uF
GND

AMS1117-3.3 from the 5 V rail: two 10 uF in and two 10 uF out per regularity.

## 1V2 LDO

U4
TLV70012-1.2
+3V3
+1V2
C6 1uF
C7 1uF
C8 4.7uF
GND

TLV70012 1.2 V core rail from 3V3. EN tied to VIN.

## FPGA

U1A
U1B
U1C
U1D
ICE40HX4K-BG121
VCCIO_0
VCCIO_1
VCCIO_2
VCCIO_3
VCCPLL0
VCCPLL1
GNDPLL0
GNDPLL1
VCC_SPI
VPP_FAST
VPP_2V5
VCC
CRESET_B
CDONE
CFG_SDO
CFG_SCK
CFG_SDI
CFG_SS_B
LED_U1_R
LED_U2_R
+3V3
+1V2
VCCPLL
GND

VPP_FAST (B10) left unconnected. Core 1.2 V, I/O banks and VPP_2V5 on 3.3 V.

## CONFIG

R4 10k CDONE
R5 10k CRESET_B
+3V3

CDONE and CRESET_B 10k pull-ups to 3V3.

## SPI FLASH

U2
W25Q128JVSIQ
R19 10k FLASH_WP_HOLD
R20 10k CFG_SS_B
CFG_SS_B
CFG_SCK
CFG_SDO
CFG_SDI
FLASH_WP_HOLD
FLASH_WP_CS
+3V3
C26 100nF
GND

Master SPI boot flash. SS_B pull-up keeps it deselected during reset; WP and HOLD pulled high.

## FPGA HEADER

J3
R11 100
R14 100
R15 100
R16 100
FA1
FA2
FA3
FA4
FA5
FA6
GND
+3V3
(no value)

4 slow user I/O (north bail row), 3V3 and GND.

## FPGA DECOUPLING

C9 100nF
C10 100nF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 100nF
C17 100nF
C18 100nF
C19 100nF
+1V2
+3V3
GND

One 100 nF per supply ball: C9–C12 core, C13–C17 bank VCCIO, C18 VCC_SPI, C19 VPP_2V5.

## PLL FILTER

FB1 (no value) +1V2 VCCPLL
C20 100nF
C21 100nF
C22 1uF
VCCPLL
GND

PLL supply: 1.2 V through a ferrite bead, 100 nF at each PLL ball, 1 uF bulk.

## LEDS

R7 1k LED_PWR D2 0603-GREEN
R8 1k CDONE LED_CDONE D3 (no value)
R9 1k LED_U1_R LED_U1 D4 0603-RED
R10 1k LED_U2_R LED_U2 D6 0603-RED
+3V3
GND

Power, CDONE and two user LEDs, 1k each: about 1.3 mA at 3.3 V.

## MOUNTING

H1 (no value)
H2 (no value)

Two M2.5 holes at opposite corners.

## TEST POINTS

TP1 +5V (no value)
TP2 +3V3 (no value)
TP3 +1V2 (no value)
TP4 (no value) CRESET_B
TP5 (no value) CDONE

Rails: 5V, 3V3, 1V2. Configuration: CRESET_B, CDONE.

Fab: jlc-advanced
77 components / 42 nets
Generated by synth-eda

Sheet: /
File: ice40_bga_node.kicad_sch

Title: ice40_bga_node
Size: A1 | Date: | Rev:
KiCad E.D.A. 10.0.0 | Id: 1/1